HDOC DAY-73

1. Create an enumeration for days (SUNDAY to SATURDAY) and print each day with its integer value.
    - Input: No Input,
    - Output: SUNDAY = 0, MONDAY = 1, TUESDAY = 2, WEDNESDAY = 3, THURSDAY = 4, FRIDAY = 5, SATURDAY = 6
2. Define an enumeration for traffic lights (RED, YELLOW, GREEN) and print 'Stop', 'Wait', or 'Go' based on its value.
    - Input: GREEN
    - Output: Go

Algorithm:

1. Start the program.
2. Define an enumeration Day containing SUNDAY to SATURDAY.
3. Since no values are explicitly assigned, C automatically assigns values starting from 0.
4. Print each day along with its integer value using printf().
5. Stop.

• C Program:

```
#include <stdio.h>
enum Day {
    SUNDAY,
    MONDAY,
    TUESDAY,
    WEDNESDAY,
    THURSDAY,
    FRIDAY,
    SATURDAY
};

int main()
```

```
{
   printf("SUNDAY = %d\n", SUNDAY);
   printf("MONDAY = %d\n", MONDAY);
   printf("TUESDAY = %d\n", TUESDAY);
   printf("WEDNESDAY = %d\n", WEDNESDAY);
   printf("THURSDAY = %d\n", THURSDAY);
   printf("FRIDAY = %d\n", FRIDAY);
   printf("SATURDAY = %d\n", SATURDAY);

   return 0;
}
```

Test Case Table:

Test Case	Input	Expected Output	Result
1	No Input	SUNDAY = 0 to SATURDAY = 6	Pass

Compilation and Execution:

The Following program has been compiled and executed successfully.

Test Case 1:



2. Enumeration for Traffic Lights

Algorithm:

1. Start the program.

2. Define an enumeration TrafficLight containing RED, YELLOW, and GREEN.
3. Read the traffic-light name from the user.
4. Compare the input with RED, YELLOW, and GREEN.
5. Assign the corresponding enum value to light.
6. Use a switch statement:
    - RED → print Stop
    - YELLOW → print Wait
    - GREEN → print Go
7. If the input is invalid, display Invalid traffic light.
8. Stop.

- **C Program:**

```
#include <stdio.h>
#include <string.h>

enum TrafficLight {
    RED,
    YELLOW,
    GREEN
};

int main()
{
    char input[10];
    enum TrafficLight light;

    printf("Enter traffic light (RED/YELLOW/GREEN): ");
    scanf("%9s", input);

    if (strcmp(input, "RED") == 0)
        light = RED;
    else if (strcmp(input, "YELLOW") == 0)
        light = YELLOW;
    else if (strcmp(input, "GREEN") == 0)
        light = GREEN;
    else {
        printf("Invalid traffic light\n");
        return 0;
```

```
    }

    switch (light)
    {
        case RED:
            printf("Stop\n");
            break;

        case YELLOW:
            printf("Wait\n");
            break;

        case GREEN:
            printf("Go\n");
            break;
    }

    return 0;
}
```

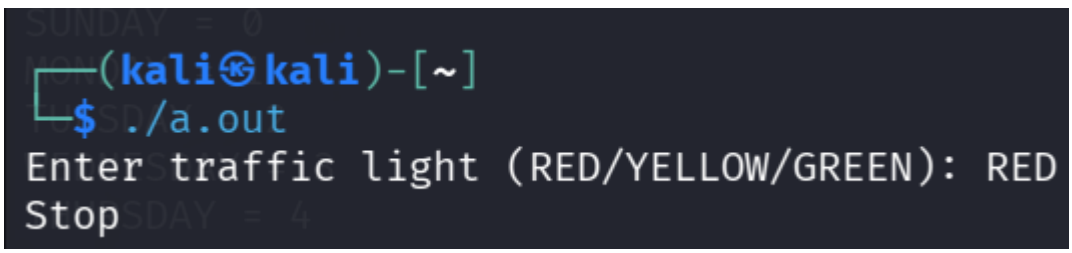
Test Case Table:

Test Case	Input	Expected Output	Result
1	RED	Stop	Pass
2	YELLOW	Wait	Pass
3	GREEN	Go	Pass
4	BLUE	Invalid traffic light	Pass

Compilation and Execution:

The following program has been compiled and executed successfully.

Test Case 1:



Test Case 2:



Test Case 3:



Test Case 4:

